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Friction stir welding systems and methods

Abstract

A friction stir welding system can reduce the likelihood of oxide formation in the work pieces being friction stir welded by forming an inert gas atmosphere in which the friction stir welding process is performed. The friction stir welding system includes one or more inert purge gas sources that direct inert purge gas towards the forming friction stir weld. The inert purge gas can be directed towards the forming weld from both an upper and a lower side, which further reduces the likelihood of oxide formation. Additionally, the friction stir welding system can include one or more housings that contains the inert purge gas about the forming friction stir weld, such as about an upper and lower side of the friction stir weld.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is a divisional application of U.S. application Ser. No. 17/386,190, filed Jul. 27, 2021, which claims priority and benefit from U.S. Provisional Application No. 63/058,341, filed Jul. 29, 2020, the contents of which are herein incorporated by reference in its entirety.

BACKGROUND
(1) Friction stir welding is one notable way for components to be joined together, particularly in industrial and manufacturing applications. Unlike traditional welding, friction stir welding does not melt the materials to join them together but instead physically intermixes the two materials together. The mixing occurs at a temperature lower than the melting point of the joined materials so that the joined materials are not unduly heated during the friction stir welding process. Excessive heating of the materials can adversely affect the properties of the materials and cause the materials to warp, form defects, or deform along the weld.
(2) Even though the materials are not heated to their melting point during the friction stir welding process, the action of mixing the materials and the heat generated during the process can cause imperfections or defects in the materials along the weld line, such as oxides or other impurities. These impurities weaken the weld and the area around it. This weakness can make the weld and the area around it susceptible to mechanical failure. In critical situations such as aerospace components, the susceptibility to mechanical failure due to imperfections or impurities in a weld is unacceptable.
(3) Conventional systems identify the imperfections or impurities caused by friction stir welding after the weld is complete. When the imperfections or impurities are found, the materials are discarded and the friction stir welding process must be redone with new materials, which is expensive and time-consuming but necessary for critical applications like welding aerospace components where human safety is paramount.
(4) Further, while oxides are formed in most types of material during the friction stir welding process, the composition of some materials like aluminum or aluminum alloys that are often used for industrial and manufacturing applications like aerospace components can increase the likelihood or impact of such oxide formation. These compositions can lead to increased brittleness of the weld and ultimately higher failure rates than other materials. In materials that are prone to oxide formation or that are adversely affected by oxide formation, the weld may be more likely to fail a quality control inspection and require re-manufacturing of the part, which is expensive and inefficient
(5) Therefore, there exists a need for friction stir welding systems and methods that minimize the likelihood of oxide formation in materials and allow materials susceptible to oxide formation to be efficiently and effectively friction stir welded.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram of an example friction stir welding system.
(2) FIG. **2**A is a partial perspective view of an example friction stir welding system.

(3) FIG. **2**B is a partial cut-away view of the example friction stir welding system of FIG. **2**A.
(4) FIG. **3** is a cutaway view of another example friction stir welding system.
(5) FIG. **4**A is a partial perspective view of another example friction stir welding system.
(6) FIG. **4**B is a partial cut-away view of the example friction stir welding system of FIG. **4**A.
(7) FIG. **5** is a process diagram of an example friction stir welding method.

DETAILED DESCRIPTION

(8) Friction stir welding systems and methods that perform a friction stir welding process in an inert gas purge atmosphere are described herein. During the friction stir welding process, two or more work pieces are joined together along a pathway that passes between the work pieces. A friction stir weld head moves along this pathway while exerting pressure against the work pieces and while rotating a pin that passes between the work pieces along the pathway. A friction stir weld backing is positioned against a side of the work pieces opposite the friction stir weld head. The frictions stir weld backing counters the pressure exerted by the friction stir weld head so that the work pieces are compressed between the friction stir weld head and the friction stir weld backing.

(9) The friction stir weld head moves along the pathway to weld the work pieces together. To form the weld, the pieces are heated by friction from the pressure exerted by the friction stir weld head and the rotation of the pin. Heating the work pieces softens or makes the work pieces more malleable but does not melt the material. As the rotating pin is moved through the work pieces along the pathway, the rotation of the pin physically intermixes the softened work pieces together to form the weld. An advantage of this friction stir welding process is that the work pieces are not heated to a melting point (which can change the material properties), additional materials are not added to the weld and the weld can be formed so that it is flush with the surfaces of the work pieces on either side of the pathway.

(10) An inert purge gas atmosphere is formed about at least a portion of the pathway by directing inert purge gas towards the pathway from an inert purge gas source. The inert purge gas atmosphere has a reduced concentration of oxygen in comparison to the normal or ambient atmosphere about the work pieces. By performing the friction stir welding process in a reduced oxygen environment, the likelihood of oxide formation in the weld is reduced. Oxides formed in the weld reduce the strength of the weld and can make the weld more prone to failure. Such a low quality weld cause safety hazards and increase the manufacturing expense. For example, a part may need to be remade multiple times to achieve the desired quality of weld or low quality welded parts may mistakenly be put into use where they present a safety hazard to users. As such, the reduction in oxide formation due to the inert gas atmosphere helps create high quality friction stir welds between work pieces.

(11) The friction stir welding systems can include multiple inert purge gas sources that direct inert purge gas towards both the upper and lower surfaces of the work pieces about where the friction stir weld is being formed by the friction stir weld head. The likelihood of oxide formation is further reduced as the portion of the work pieces being welded is substantially surrounded by inert purge gas because the inert gas forces oxygen away from the welded surfaces and reduces the likelihood that oxides can form along those surfaces at various points along the weld. Alternatively, the inert purge gas atmosphere can be formed around one of the upper and lower surfaces of the work pieces about where the friction stir weld is being formed. For example, the an inert purge gas source can direct inert purge gas towards the friction stir weld head where the work pieces are more likely to experience oxide formation as a result of the friction stir welding process.

(12) The inert purge gas source can include various outlets to direct or contain the inert purge gas about the pathway where the friction stir weld is being formed. For example, the inert purge gas source can include a nozzle that directs inert purge gas towards the friction stir weld head and the forming friction stir weld. In another example, a housing can be used to contain the inert purge gas about the friction stir weld. The inert gas can be directed both above and below the friction stir weld, both on the same and the opposing side from the weld head. The inert gas source can be a

single source or could include multiple sources or multiple outlets from a single source, each directed towards the weld head, the friction stir weld, or both. In an embodiment in which multiple inert purge gas sources direct inert purge gas both above and below the weld pathway (on the same side and the opposing side from the weld head), a housing can be included about the upper and lower portions to optimize the inert purge gas atmosphere around the weld.

(13) The inert purge gas source can be stationary or could move with the friction stir weld head during the friction stir welding process. To maintain the inert purge gas atmosphere about the forming friction stir weld, the inert purge gas source can include a carriage that moves the inert purge gas source along the pathway so that the inert purge gas atmosphere is maintained where the friction stir weld is being formed. Alternatively, the friction stir weld head can include a carriage to which the inert purge gas source can be coupled so that the inert purge gas source is moved with the friction stir weld head during the friction stir welding process. In a further alternative, the inert purge gas source can be substantially stationary relative to the friction stir weld head. For example, the inert purge gas source can be arranged about a portion of the pathway and can direct inert purge gas about this portion of the pathway. The friction stir weld head can traverse the portion of the pathway while the inert purge gas is being dispensed so that the friction stir weld formed along the portion of the pathway is formed in an inert purge gas atmosphere. The inert purge gas source can extend along the entire length of the pathway to form the inert purge gas atmosphere, or the inert purge gas source can be moved to a subsequent portion of the pathway so that the inert purge gas atmosphere is maintained as the friction stir weld head traverses along the pathway.

(14) Depending on the materials of the work pieces, such as their physical and chemical properties, various parameters of the friction stir welding process can be varied. Example parameters can include a rotational speed of the friction stir weld head and a traverse speed of the friction stir weld head along the pathway between the work pieces. The materials can have tolerances associated with the various parameters of the friction stir welding process, such as a range of appropriate rotational and traverse speeds that form a high quality weld between the work pieces. For example, some materials may have a wide range of rotational and traverse speeds that can be used when friction stir welding and other materials may have narrower acceptable ranges of rotational and traverse speeds that can be used to achieve a similar quality of friction stir weld. For materials having smaller or reduced tolerances for the friction stir welding process, the use of the inert gas atmosphere can expand the operating tolerances while maintaining the quality of the weld. Since performing the friction stir weld in an inert gas atmosphere can reduce imperfections, such as oxide formation, in the weld, the operating parameters of the friction stir welding process can be expanded while maintaining a similar or reduced level of imperfections with respect to forming the weld in a normal environment. The expansion of the operating parameters can reduce the precision needed during the friction stir welding process and can allow for a higher output of friction stir welding, such as by increasing the allowable traverse speed of the friction stir weld head. In this manner, the efficiency of the friction stir welding process can be increased due to performing the friction stir welding process in an inert gas atmosphere.

(15) Additionally, some materials may have such a reduced or low amount of workability, i.e. such low tolerances or such a high likelihood of oxide formation, that a normal friction stir welding process is not able to create welds of consistent or acceptable quality. The use of the inert gas atmosphere can allow such materials to be friction stir welded. For example, aluminum alloys having a high lithium content, such as 0.5% or greater lithium content, can have desired properties for aerospace applications like building rockets that require critical levels of safety against component failures. Aluminum alloys are attractive in building such components because they offer a high strength to weight ratio, but they have a low workability that makes them unsuitable for friction stir welding. Using friction stir welding systems and methods described herein, high lithium content, aluminum alloys can be successfully friction stir welded together in a consistent and high quality manner. Similarly, other materials having low workability can also be successfully

joined by performing the friction stir welding process in an inert gas atmosphere, as described below.
(16) FIG. **1** illustrates an example friction stir welding system **100** that can be used to weld work pieces **110** together along pathway **119** to form a weld **118** having reduced oxide formation. An upper inert gas source **140** and, optionally, a lower inert gas source **160** supply inert gas about the pathway **119** of the weld **118** to provide a reduced oxygen atmosphere in which the frictions stir welding process occurs. The reduced oxygen about the pathway **119** reduces or prevents the formation of oxides along the pathway **119** due to the mechanical action and heat generated by the friction stir welding process. Additionally, performing the friction stir welding process in the reduced oxygen environment can expand the operating parameters of the friction stir welding process, reducing the complexity of the process and increasing the efficiency of the process.
(17) The work pieces **110** are the pieces of material **112** that are to be welded **118** together along the pathway **119**. The work pieces **110** are arranged so that they contact or about each other along respective sides of the work pieces **110** and the point of contact is the pathway **119** between the work pieces **110**. The friction stir welding process occurs along the pathway **119** to form the weld **118** that joins the work pieces **110** together. To maintain contact between the work pieces **110**, the work pieces **110** can be placed in a jig or fixture that orients and arranges the work pieces **110** properly for the friction stir welding process.
(18) The material **112** of the work pieces **110** can effect various parameters of the friction stir welding process. For example, the chemical composition of the material **112**, the mechanical properties of the material **112**, other attributes of the material **112**, or combinations thereof, can be used to determine a rotational speed, a traverse speed, or other parameters of the friction stir welding process. Additionally, the tolerances of these parameters, i.e. how much the parameter can vary from a set amount, can be dependent on the materials **112**. Certain materials **112** may have large tolerances, such as allowing a wide range of rotational and traverse speeds to be used during the friction stir welding process while maintaining a desired or required quality of the weld **118**. Other materials can have tighter tolerances, such as allowing a narrow range of rotational and traverse speeds to be used during the friction stir welding process while maintaining a desired or required quality of the weld **118**. The quality of the weld **118** can include a tolerated amount of defects or impurities in the weld, as these may weaken the weld **118**.
(19) A friction stir weld head **120** and a friction stir weld backing **130** are positioned on opposing sides of the work pieces **110** and engage the work pieces **110**. The friction stir weld head **120** engages first or upper surfaces of the work pieces **110** and the friction stir weld backing **130** engages second or lower surfaces of the work pieces **110**. In this manner, the work pieces **110** are positioned between the friction stir weld head **120** and the friction stir weld backing **130**. The friction stir weld head **120** and the friction stir weld backing **130** are positioned to align with the pathway **119**, where the work pieces **110** abut or contact each other. During the friction stir welding process, the friction stir weld head **120** and, optionally, the friction stir weld backing **130** move along the pathway **119** to form the weld **118** that joins the work pieces **110** along the pathway **119**.
(20) The friction stir weld head **120** includes an upper shoulder **122** and a pin **124** that are rotated during the friction stir welding process. The upper shoulder **122** is a portion of the friction stir weld head **120** that contacts and exerts pressure on the upper surfaces of the work pieces **110**. The exertion of pressure on the upper surfaces of the work pieces **110** compresses the lower surfaces of the work pieces **110** against the friction stir weld backing **130**. This compression of the work pieces **110** assists with generating the friction of the frictions stir welding process. Additionally, this compression prevents deformation of the materials along the weld **118**, creating a weld **118** that is flush with the upper and lower surface of the work pieces **110**.
(21) The upper shoulder **122** is circularly shaped and has a diameter. The diameter of the upper shoulder **122** extends across portions of the work pieces **110** on either side of the pathway **119** of the weld **118**. During the friction stir welding process, the upper shoulder **122** is translated along

the pathway **119** and across the upper surfaces of the work pieces **110**, which compresses the work pieces **110** against the friction stir weld backing **130**. The pressure exerted by the upper shoulder **122** on the work pieces **110** can be varied, such as depending on the material **112** being joined, desired mechanical properties of the weld **118** between the work pieces **110**, other considerations, or combinations thereof.

(22) A pin **124** of the friction stir weld head **120** is positioned concentric to the upper shoulder **122** and extends a length from the upper shoulder **122**. The length the pin **124** extends from the upper shoulder **122** is the depth to which the pin **124** will engage the work pieces **110**. The pin can be releasably coupled to the friction stir weld head **120** to allow the length and shape of the pin **124** to be varied. Varying the length of the pin **124** changes the depth to which the pin **124** engages the work pieces **110**. This can alter various properties of the weld **118** and the length of the pin **124** can be selected based on the materials **112**, the desired properties of the weld **188**, other considerations, or combinations thereof. For example, the pin **124** can extend partially through the thickness of the work pieces **110**, which welds **118** a portion of the thickness of the work pieces **110** along the pathway **119**. In another example, the pin **124** can extend substantially through the thickness of the work pieces **110**, which welds **118** substantially the entire thickness of the work pieces **110** along the pathway **119**. As previously mentions, the pin **124** has a shape which can also effect various properties of the weld **188**. In an example, the pin **124** can have a cylindrical, conical or other shape. The shape of the pin **124** can be selected based on various considerations, such as the material **112** of the work pieces **110**, the desired properties of the weld **118**, other considerations, or combinations thereof.

(23) In the example shown in FIG. **1**, the friction stir weld backing **130** is positioned against the work pieces **110**, opposing the friction stir weld head **120**. The friction stir weld backing **130** can counter the pressure exerted on the work pieces **110** by the friction stir weld head **120** or can assist with compressing the work pieces **110** between the friction stir weld head **120** and the friction stir weld backing **120**. In an example, the friction stir weld backing **130** can be a lower shoulder **132**. The lower shoulder **132** can be coupled to the friction stir weld head **120** so that it moves with the friction stir weld head **120** as it is translated along the pathway **119**. To couple the lower shoulder **132** to the friction stir weld head **120**, the lower shoulder **132** can be coupled to the pin **124** that extends through the thickness of the work pieces **110**. The lower shoulder **132** can rotate with the pin **124** during the friction stir welding process due to the coupled nature of the lower shoulder **132** and the pin **124**. Alternatively, a portion of the lower shoulder **132** can rotate with the pin **124** during the friction stir welding process while another portion of the lower shoulder **132** does not rotate with the pin **124**. The coupling between the friction stir weld head **120** and the lower shoulder **132** can be adjustable to allow the compression of the work pieces **110** to be varied. The compression of the work pieces **110** can be varied based on a variety of considerations, such as the materials **112** being joined, the properties of the weld **118**, other considerations, or combinations thereof.

(24) In another example, the friction stir weld backing **130** can be a backing surface **134**. The backing surface **134** can remain stationary during the friction stir welding process and can support the lower surfaces of the work pieces **110** along the pathway **118**. The friction stir weld head **120** can exert pressure on the work pieces **110**, compressing them between the friction stir weld head **120** and the backing surface **134**. Unlike the lower shoulder **132**, the backing surface **134** does not need to be coupled to the friction stir weld head **120**, so the pin **124** does not need to extend through the thickness of the work pieces **110**. This can allow the weld **118** to be a partial thickness weld that does not extend through the entire thickness of the work pieces **110**. Alternatively, the backing surface **134** can be used with a pin **124** that extends substantially through the thickness of the work pieces **110** to create a full thickness weld **118** that extends from the upper surfaces of the work pieces **110** to the lower surfaces of the work pieces **110**, or substantially thereto.

(25) During the friction stir welding process, the upper shoulder **120** and the pin **124** of the friction

stir weld head **120** are rotated and the rotating upper shoulder **12** and pin **124** are translated along the pathway **119** to form the weld **118** that joins work pieces **110** together. The compression of the work pieces **110** between the friction stir weld head **120** and the friction stir weld backing **130** generates friction, which heats the work pieces **110** about where the friction stir weld head **120** contacts the work pieces. The generated heat from the applied friction softens, but does not melt the material **112** of the work pieces **110**. No other heat source is applied to the materials during the welding process. The rotating pin **124** moves along the pathway **119** where the work pieces **110** contact each other. As the rotating pin **124** moves through the softened material **112** of the work pieces **110**, the rotation of the pin **124** causes the material **112** of the work pieces **110** to be intermixed. This intermixing physically mixes the material **112** of the work pieces **110** together about the pathway **119**, which forms the weld **118** that joins the work pieces **110**. As the friction stir weld head **120** passes through the work pieces **110** and intermixes them along the pathway **119**, the formed weld **118** quickly cools since the friction stir weld head **120** is no longer contacting that portion of the work pieces **110**. This quick cooling limits deformation of the work pieces **110** along the weld **118** and results in a butt weld that is flush with the upper and lower surfaces of the work pieces **110**.

(26) As part of the parameters of the friction stir welding process, the speed of rotation of the pin **124**, the speed of translation of the pin **124** and upper shoulder **122** along the pathway **119** and the compression of the work pieces **110** along the pathway **119** can be varied. Various parameters, such as these, can be altered or changed depending on factors, such as the materials **112** being joined, the desired mechanical properties of the weld **118**, a desired quality of the weld **118**, other considerations, or combinations thereof.

(27) Certain materials **112** may be more prone to form defects or impurities, such as the formation of oxides, in the weld **118** during the friction stir welding process. Such defects or impurities can create a weak weld **118** that may be prone to failure, such as forming brittle welds or materials surrounding the welds that crack and break easier than the material otherwise would fail without the defects. In an example, high lithium content aluminum-lithium (Al—Li) alloys **114** can be prone to form oxides and are conventionally unsuitable for friction stir welding as there is a high likelihood the weld **118** will have such defects and be prone to failure. The benefit of high lithium content Al—Li alloys **114**, such as alloys containing at least 0.5% lithium, is that they have desirable properties, such as having a high strength to weight ratio. Example Al—Li alloys **114** include AL 2195 **116**, AL 2050, AL 2090. However, the high lithium content of the Al—Li alloys **114** causes the alloys to have a low workability making them difficult to join together and increasing a likelihood of forming a low quality weld **118** due to the formation of oxides in the weld **118**. The friction stir welding system **100** includes inert gas sources **140** and **160** that form an inert gas atmosphere about the pathway **119** where the weld **118** is being formed by the friction stir weld head **120**. The inert gas atmosphere has a reduced level of oxygen which reduces the formation of oxides in the weld **118**, allowing high lithium content Al—Li alloys **114**, such as AL 2195 **116**, to be joined using a friction stir welding process that forms a high quality weld **118**.

(28) As discussed, materials **112** may have varying tolerances to the rotational speed of the pin **124** and upper shoulder **122** and the traverse speed of the friction stir weld head **120** along the pathway **119**. The inert gas atmosphere about weld **118** being formed by the friction stir weld head **120** can expand these operating tolerances while maintaining the same quality of the weld **118**. For example, a certain material **112** may have a fairly narrow range of suitable traverse speed at which the friction stir weld head **120** may be moved along the pathway **119**, such as ±2 mm/min. Using the inert gas atmosphere, the suitable range of the traverse speed can be ±8 mm/min, while maintaining a weld **118** quality that is at least similar to the suitable traverse speed in a normal atmosphere. The expanded operating tolerances of the friction stir welding process using the inert gas atmosphere can increase the speed and reduce the complexity or precision of the friction stir welding process.

(29) To form the inert gas atmosphere, an inert purge gas **150**, such as argon **152**, can be dispensed about the pathway **119** where the weld **118** is being formed by the friction stir weld head **120**. The inert purge gas **150** is a non-reactive gas that has a greater density than the normal atmosphere so that it displaces the normal atmosphere about the pathway **119**. Additionally, the inert gas atmosphere remains about the pathway **119** for a period of time since the density of the inert purge gas **150** slows its dissipation from about the pathway **119**. For example, argon **152** has a high density and low reactivity that allow it to form an inert gas atmosphere about the pathway **119**. Other gases having a high density and low reactivity, such as nitrogen, carbon dioxide, can be used to form the inert gas atmosphere about the pathway **119**. The selection of the inert purge gas **150** can also be dependent on the materials **112** being used and their reactivity with the inert purge gas **150**. A combination of inert purge gas **150** and materials **112** having a low reactivity assists in minimizing the formation of defects or impurities in the weld **118**.
(30) The inert gas atmosphere about the pathway **119** can be formed by dispensing the inert purge gas **150** from one or both of the upper inert purge gas source **140** and the lower inert purge gas source **160**. The upper inert purge gas source **140** dispenses the inert purge gas **150** about the upper surfaces of the work pieces **110** along the pathway **119**. Similarly, the lower inert purge gas source **160** dispenses the inert purge gas **150** about the lower surfaces of the work pieces **110** along the pathway **119**. In this arrangement, the weld **118** is formed in an inert gas atmosphere which assists in creating a high quality weld **118** and assisting with minimizing the formation of impurities in the weld **118**.
(31) The upper inert purge gas source **140** is in fluid communication with a source of the inert purge gas **150**. Inert purge gas **150** from the inert purge gas source is directed through an outlet **142** of the upper inert purge gas source **140**. The inert purge gas **150** flows from the outlet **142** and about at least a portion of the weld **118**, a portion of the pathway **119**, another portion of the friction stir welding system **100**, or a combination thereof. The outlet can be positioned near the friction stir weld head **120** and can be oriented to direct the inert purge gas **150** towards the weld **118**, a the pathway **119**, another portion of the friction stir welding system **100**, or a combination thereof. For example, the outlet **142** can be positioned away from the upper surfaces of the work pieces **110** and at an angle relative to the work pieces **110**, such as angled downwards towards weld **118** or pathway **119**. In another example, the outlet **142** can be angled relative to the friction stir weld head **120**, such as to direct inert purge gas towards pathway **119** ahead of the friction stir weld head **120** or to direct inert purge gas towards the formed weld **118** immediately behind the friction stir weld head **120**. Alternatively, the outlet **142** can be angled relative to both the work pieces **110** and the friction stir weld head **120**.
(32) The size and profile of the outlet **142** can be selected to assist with dispensing the inert purge gas **150** from the upper inert purge gas source **140**. For example, the outlet **142** can be sized so that a desired or required amount of inert purge gas **150** per unit time is dispensed from the outlet, the outlet **142** can be shaped to prevent mixing of the dispensed inert purge gas **150** with the surrounding environment, other considerations regarding the dispensing of the inert purge gas **150** from the upper inert purge gas source **140**, or combinations thereof. Additionally, the upper inert purge gas source **140** can include a flow regulating device or system to control the rate at which inert purge gas **150** is dispensed through the outlet **142**. The flow rate of the inert purge gas **150** from the upper inert purge gas source **140** can be preset before the friction stir welding process begins or can vary as the friction stir welding process progresses. In this manner, the flow of the inert purge gas **150** can be controlled to assist in maintaining the inert purge gas atmosphere during the friction stir welding process.
(33) An example outlet **142** of the upper inert purge gas source **140** can include a nozzle **143** having one or more openings through which inert purge gas **150** can be dispensed. The nozzle **143** can be fixedly or adjustably mounted. In the fixed embodiment, the nozzle **143** can have a fixed position and orientation, such as relative to the pathway **119** or friction stir weld head **120**. In the

adjustably mounted embodiment, the position, orientation, or both of the nozzle **143** can be adjusted. This can allow the nozzle **143** to be positioned closer or further from the pathway **119**, the upper shoulder **122**, the upper surfaces of the work pieces **110**, or other portions of the friction stir welding system **100**. Additionally, the orientation of the nozzle **143** can be adjustable to allow the dispensed inert purge gas **150** to be directed in a controlled manner. For example, the nozzle **143** can be oriented so that the inert purge gas **150** is dispensed towards the leading side of the friction stir weld head **120**. In this manner, the inert purge gas atmosphere can be formed slightly ahead of the friction stir weld head **120** so that the friction stir welding performed by the upper shoulder **122** and pin **124** occurs in the inert purge gas atmosphere. The adjustable nozzle **143** can be manually adjustable or adjustable automatically or remotely. This can allow the position, orientation, or both of the nozzle **143** to be varied during the friction stir welding process. The variable adjustability may be desirable during friction stir welding processes that occur along a curved pathway **119**, so that the inert purge gas atmosphere can be more efficiently formed about the forming weld **118**.
(34) In another example, the outlet **142** of the upper inert purge gas source **140** can include a perforated tubing **144**. The perforated tubing **144** can be circular tubing that has a series of openings disposed only its length. The inert purge gas **150** can be dispensed through the openings of the perforated tubing **144** and the perforated tubing **144** can be positioned and oriented to generate the inert purge gas atmosphere for the friction stir welding process. For example, the perforated tubing **144** can be placed along the upper surface of a work piece **110** so that the perforated tubing **144** is proximal or near the pathway **119**. The perforated tubing **144** can be oriented to that the openings of the perforated tubing **144** direct or dispense the inert purge gas **150** towards the pathway **119** to create an inert purge gas atmosphere about the forming weld **118**.
(35) The perforated tubing **144** can be flexible or rigid. Flexible perforated tubing **144** can be arranged on or near the work pieces **110** to follow the pathway **119** so that the weld **118** formed by the friction stir weld head **120** is formed in an inert purge gas atmosphere. A user can arrange the flexible perforated tubing **144** prior to the friction stir welding process or can move the perforated tubing **144** as the friction stir welding process is ongoing so that the inert purge gas atmosphere is maintained about the forming weld **118**. Rigid perforated tubing **144** can be similarly arranged but may require bending or joining to follow a non-linear pathway **119**. In an example, the work pieces **110** can be held in a jig or fixture during the friction stir welding process and the perforated tubing **144** can be integrated with or coupled to the jig or fixture.
(36) As discussed, the inert purge gas **150** flows through the openings of the perforated tubing **144** to form the inert purge gas atmosphere. The perforated tubing **144** can include flow control features, such as baffles, to assist with maintaining a substantially constant flow rate of inert purge gas **150** from the openings of the perforated tubing **144**. Additionally, the size and shape of the openings of the perforated tubing **144** can be varied along the length of the perforated tubing **144** to assist with maintaining a steady flow rate of inert purge gas **150** from the perforated tubing **144**.
(37) In another example, the outlet **142** can include a housing **145** that assists with containing the inert purge gas **150** about an area. For example, the housing **145** can be positioned about the pathway **119** and inert purge gas **150** can be dispensed or flowed into the housing **145** to form the inert purge gas atmosphere. The friction stir weld head **120** can then move through the housing as it traverses along the pathway **119**, so that the weld **118** is formed in the inert purge gas atmosphere. In an embodiment, the housing **145** can be a temporary structure, such as made of a frangible material that breaks apart as the friction stir weld head **120** traverses along the pathway **119**. As the friction stir weld head **120** traverses along the pathway **119**, the temporary housing **145** does not impede it and is instead destroyed or moved to allow the friction stir weld head **120** to progress. Also, during the friction stir welding process, the inert purge gas **150** can be flowed continuously into the housing **145** to maintain the inert purge gas atmosphere within the housing **145** and about the forming weld **118**.
(38) The upper inert purge gas source **140** can also include a carriage **146** that can move the upper

inert purge gas source **140** during the friction stir welding process. The carriage **146** can be a device or system that moves the upper inert purge gas source **140**, such as along the pathway **119** during the friction stir welding process to assist with maintaining the inert purge gas atmosphere about the forming weld **118**. The carriage **146** can move or be moved as the friction stir welding process is occurring so that the upper inert purge gas source **140** is similarly moved. In this manner, the upper inert purge gas source **140** can move in a synchronized manner with the friction stir weld head **120** to maintain the inert purge gas atmosphere.
(39) In an embodiment, a user can move the carriage **146** or can cause the carriage **146** to be moved, such as by using a control system. The user can reposition the carriage **146** at intervals so the upper inert purge gas source **140** is positioned to form the inert purge gas atmosphere about the forming weld **118**. For example, the upper inert purge gas source **140** can have an outlet **142** that is perforated tubing **144** having a first length. When the friction stir weld head **120** is near a first end of the perforated tubing **144**, the user can move the carriage **146** so that a second end of the perforated tubing **144** is then near the friction stir weld head **120**. As the friction stir weld head **120** continues traversing along the pathway **119**, the user can repeat the process to move the carriage **146** and reposition the perforated tubing **144**. Alternatively, the user upper inert purge gas source **140** does not include the carriage **146** and the user can manually reposition the perforated tubing **144** as needed.
(40) In another embodiment, the movement of the carriage **146** can be controlled by a control system that can move the carriage **146** as needed. For example, the carriage **146** can be a robotic arm that moves the upper inert purge gas source **140** or a portion thereof, such as the outlet **142**. Alternatively, the upper inert purge gas source **140** or a portion thereof, can be mounted to the robotic arm so that it moves with the robotic arm. As the friction stir welding process occurs, the robotic arm can reposition or move the upper inert purge gas source **140**, or a portion thereof, to maintain the inert purge gas atmosphere about the forming weld **188**.
(41) Alternatively, the friction stir weld head **120** can include an inert purge gas carriage **126** that moves the upper inert purge gas source **140**, or a portion thereof. In this manner, the upper inert purge gas source **140**, or a portion thereof, is moved with the friction stir weld head **120** as it traverses along the pathway **119**. Since the upper inert purge gas source **140** is moved with the friction stir weld head **120**, the inert purge gas atmosphere can be maintained about the forming weld **118**. The inert purge gas carriage **126** can be an interface of the friction stir weld head **120** to which the upper inert purge gas source **140**, or a portion thereof, can be mounted.
(42) The friction stir welding system **100** can also include a lower inert purge gas source **160** that assists in forming an inert purge gas atmosphere about pathway **119** on the lower surfaces of the work pieces **110**. Like the upper inert purge gas source **140**, the lower inert purge gas source **160** can include an outlet **162**, such as a nozzle **163**, perforated tubing **164** or a housing **165**, to direct the inert purge gas **150** towards the pathway **119**. The lower inert purge gas source **160** can also include a carriage **166** to move the lower inert purge gas source **160**, or a portion thereof, along the pathway **119** in a manner similar to the carriage **146** of the upper inert purge gas source **140**. Similarly, the friction stir weld backing **130** can include an inert purge gas carriage **136** to move the lower inert purge gas source **160**, or a portion thereof, instead the carriage **166** of the lower inert purge gas source **160**.
(43) Additionally, the friction stir welding system **100** can include multiple upper inert purge gas sources **140-140***n*, multiple lower inert purge gas sources **160-160***n*, or a combination thereof. The use of multiple inert purge gas sources **140-140***n*, **160-160***n* can assist with forming the inert purge gas atmosphere in which the friction stir welding process occurs to minimize or prevent the formation of oxides in the material **112** of the work pieces **110**. In an example, the friction stir welding system **100** can include a multiple upper inert purge gas sources **140-140***n*. A first upper inert purge gas source can be positioned and oriented to dispense or direct inert purge gas **150** towards a leading portion of the friction stir weld head **120**. A second upper inter purge gas source

can be positioned and oriented to dispense or direct inert purge gas **150** towards a retreating portion of the friction stir weld head **120**. In this manner, inert purge gas **150** is dispensed preceding the formation of the weld **118** and after the weld **118** has been formed. This can form an inert purge gas atmosphere in which the weld **118** is formed and cooled, which can assist with minimizing or preventing the formation of oxides in the material **112** of the work pieces **110**. Alternatively, the multiple inert purge gas sources **140-140***n*, **160-160***n* can be positioned proximal the work pieces **110** and oriented to direct inert gas towards the pathway **119**.

(44) For example, a first upper inert purge gas source **140** can be positioned away from an upper surface of a first work piece **110** and angled to direct the inert purge gas **150** towards the pathway **119**. A second upper inert purge gas source **140***n* can be positioned away from an upper surface of a second work piece **110** and also angled to direct the inert purge gas **150** towards the pathway **119**. A similar arrangement of multiple lower inert purge gas sources **160-160***n* can also be used. In these arrangements, multiple inert purge gas sources **140-140***n*, **160-160***n* can be positioned on opposing sides of the pathway **119** direct inert purge gas **150** towards the pathway **119**. Alternatively, the multiple inert purge gas sources **140-140***n*, **160-160***n* can be positioned on the same side of the pathway **119** and direct inert purge gas towards the same portion of the pathway **119** or weld **118**, or towards different portions of the pathway **119** or weld **118**. Additionally, the angling of the inert purge gas sources **140-140***n*, **160-160***n* relative to the work pieces **110** can be the same or different. For example, a first inert purge gas source **140**, **160** can be angled at a first angle relative to the work pieces **110** and a second inert purge gas source **140***n*, **160***n* can be angled at a second angle relative to the work pieces **110**. Alternatively, the multiple inert purge gas sources **140-140***n*, **160-160***n* can be angled at substantially the same angle relative to the work pieces **110**.

(45) FIG. **2**A is a perspective view of an example friction stir welding system **200** and FIG. **2**B is a partial cut-away of the friction stir welding system **200**. The friction stir welding system **200** includes a friction stir weld head **120** and a friction stir weld backing **130** that are welding two work pieces **110***a*, **110***b* together along a pathway **119**. The friction stir welding system **200** also includes two upper inert purge gas sources that have perforated tubing **145***a*, **145***b* outlets and a lower inert purge gas source that has perforated tugging **165**. Inert purge gas **150** is dispensed or directed from the perforated tubing **145***a*, **145***b*, **165** to form an inert purge gas atmosphere about the pathway **119**. The friction stir welding process by the friction stir weld head **120** and friction stir weld backing **130** occurs in the inert purge gas atmosphere, which assists in minimizing or preventing the formation of oxides in the materials of the work pieces **110***a*, **110***b*, such as in a weld **118**.

(46) The friction stir weld head **120** includes an upper shoulder **122** that contacts upper surfaces **202***a* and **202***b* of the work pieces **110***a*, **110***b*. A pin **124** of the frictions stir weld head **120** extends through the thickness of the work pieces **110***a*, **110***b*, from the upper surfaces **202***a*, **202***b* to the lower surfaces **203***a*, **203***b*. A lower shoulder **132** of the friction stir weld backing **130** is coupled to the pin **124** and contacts the lower surfaces **203***a*, **203***b*. During the friction stir welding process, the friction stir weld head **120** exerts pressure on the upper surfaces **202***a*, **202***b* and the friction stir weld backing **130** counters the exerted pressure. Also, during the friction stir welding process, the friction stir weld head **120** rotates and the combination of the pressure on and rotation against the work pieces **110***a*, **110***b* generates friction that heats the work pieces **110***a*, **110***b* below the melting temperature of the materials. The rotation physically mixes portions of the work pieces **110***a*, **110***b* together, forming the weld **118**. The friction stir weld head **120** and coupled friction stir weld backing **130** traverse **206** across the work pieces **110***a*, **110***b* along the pathway **119**, which joins the work pieces **110***a*, **110***b* together.

(47) To assist with forming a high quality weld **118**, such as by minimizing or preventing defects like oxide formation, an inert purge gas atmosphere is formed about the stir weld head **120** and friction stir weld backing **130**. The perforated tubing **145***a* and **145***b* are arranged on the upper surfaces **202***a*, **202***b* of the work pieces **110***a*, **110***b* and are positioned near the pathway **119**.

Openings **210** of the perforated tubing **145***a*, **145***b* are oriented towards the pathway **119** so that inert purge gas is directed towards the pathway **119** and the friction stir weld head **120** to form the inert purge gas atmosphere. The openings **210** can be spaced regularly along the length of the perforated tubing **145***a*, **145***b* to direct the inert purge gas evenly along the pathway **119**. Alternatively, the spacing, size, shape, or combinations thereof, of the openings **210** can be varied to control the flow of inert purge gas from the perforated tubing **145***a*, **145***b*.
(48) The perforated tubing **165** is positioned below the work pieces **110***a*, **110***b* near the lower surfaces **203***a*, **203***b*. The openings **210** of the perforated tubing **165** are similarly oriented towards the pathway **119** to assist in forming the inert purge gas atmosphere about the pathway **119**. In the example shown in FIGS. **2**A-**2**B, the openings **210** of the perforated tubing **165** direct inert purge gas about the friction stir weld backing **130** and form an inert purge gas atmosphere about the lower surfaces **203***a*, **203***b* of the work pieces **110***a*, **110***b*.
(49) In the example shown in FIGS. **2**A-**2**B, the perforated tubing **145***a*, **145***b*, **165** is shown arranged along the pathway **119**. The perforated tubing **145***a*, **145***b*, **165** can extend along the length of the pathway **119**, as shown, or the perforated tubing **145***a*, **145***b*, **165** can be segments or sections of perforated tubing that can be moved during the friction stir welding process. For example, the perforated tubing **145***a*, **145***b*, **165** can be positioned to extend along a portion of the pathway **119** that is being friction stir welded. When the friction stir welding process nears the end of the portion of the pathway **119**, the perforated tubing **145***a*, **145***b*, **165** can be moved along the pathway **119** to be positioned along a subsequent section of the pathway **119** along which the friction stir welding process will continue. So instead of having the perforated tubing **145***a*, **145***b*, **165** extending along the entire length of the pathway **119**, the perforated tubing **145***a*, **145***b*, **165** can be repositioned as needed to maintain the inert purge gas atmosphere about where the weld **118** is being formed. The perforated tubing **145***a*, **145***b*, **165** can be moved and repositioned manually, such as by a user, or by an automated or controlled process, such as using a carriage. For example, the perforated tubing **145***a*, **145***b*, **165** segments can be connected to the friction stir weld head **120** and the friction stir weld backing **130**, respectively. As the friction stir weld head **120** and friction stir weld backing **130** traverse **206** along the pathway **119**, the segments of perforated tubing **145***a*, **145***b*, **165** can be similarly moved along the work pieces **110***a*, **110***b*. In this manner, the inert purge gas atmosphere created by the perforated tubing **145***a*, **145***b*, **165** also moves with the friction stir welding process as it traverses **206** along the pathway **119**.
(50) FIG. **3** is a cut-away view of an example friction stir welding system **300**. In this example, the view is taken looking at the side of a work piece **110** along which a friction stir weld head **120** traverses **206**. Similar to the examples of FIGS. **1**, and **2**A-**2**B, the friction stir weld head **120** includes an upper shoulder **112** that contacts an upper surface **202** of the work piece **110** and a pin **124** that extends through the work piece **110**. The pin **124** is coupled to a friction stir weld backing **130** that includes lower shoulder **132** that contacts a lower surface **203** of the work piece **110**. The work piece **110** is compressed between the friction stir weld head **120** and the friction stir weld backing **130**. Additionally, the friction stir weld head **120** and backing **130** rotate. The combination of the pressure and rotation generates friction with the work piece **110**, heating the work piece **110** along the pathway. The softened material of the work piece **110** is “stirred” by the pin **124** with similar material of another work piece (not shown) to weld the two work pieces together. In the example shown in FIG. **3**, the pin **124** has a conical shape with fluting. Alternatively, as discussed above, the pin **124** can have other profiles and features that effect how the weld is formed between two work pieces. The characteristics of the pin **124** can be selected based on a variety of factors, such as the material of the work pieces.
(51) An upper inert gas source **140** is positioned near a leading edge of the friction stir weld head **120** and dispenses inert urge gas **150** ahead of the friction stir weld head **120**. In this manner, the friction stir weld head **120** traverses **206** into the inert purge gas **150** dispensed from the upper inert purge gas source **140**. The upper inert purge gas source **140** can include a diffuser **302** that can

slow and distribute the inert purge gas **150** as it is dispensed from the upper inert purge gas source **140**. The diffuser **302** can be a membrane or element that has multiple openings that regulate the flow of inert purge gas through the diffuser **302**. The regulation of the flow of inert purge gas **150** through the diffuser can slow the flow of the inert purge gas **150**. Dispensing the inert purge gas **150** in a slow manner can prevent mixing of the inert purge gas **150** with the surrounding environment and keep the inert purge gas **150** in a more contained area as it is not deflected away from where it is dispensed due to the speed of the exiting inert purge gas **150**. The upper inert purge gas source **140** can move independently from the friction stir weld head **120** as it traverses **206** along the work piece **110**, or the upper inert purge gas source **140** can be coupled to the friction stir weld head **120** so they move as a single unit. By coupling the upper inert purge gas source **140** and the friction stir weld head **120** together, the inert purge gas **150** is consistently dispensed in front of the friction stir weld head **120** so that the friction stir welding by the upper shoulder **122** and pin **124** occurs in an inert purge gas atmosphere. Performing the friction stir welding process in an inert purge gas atmosphere assists in minimizing or preventing the formation of oxides.

(52) A lower inert purge gas source **160** is positioned near the friction stir weld backing **130**. The lower inert purge gas source **160** includes a nozzle **163** from which inert purge gas **150** is directed towards the lower surface **203** of the work piece **110** to assist in forming an inert purge gas atmosphere. In the example shown in FIG. **3**, the inert purge gas **150** is dispensed behind the direction in which the weld is being formed, i.e. the direction of traverse **206**. However, the rotation of the friction stir weld backing **130** can draw the inert purge gas **150** about where the weld is being formed. Alternatively, the positioning of the lower inert purge gas source **160** can be altered to change the are about which the inert purge gas **150** is dispensed or directed about the lower surface **203** of the work piece **110**.

(53) The lower inert purge gas source **160** is connected to a carriage **136** of the friction stir weld backing **130**. The carriage **136** includes arms **312***a*, **312***b* to which the lower inert purge gas source **160** can be coupled. The arms **312***a*, **312***b* extend from a ball bearing **310**, which prevents the arms **312***a*, **312***b* from being rotated by the rotation of the friction stir weld backing **130**. The ball bearing **310** allows the friction stir weld backing **130** to rotate an inner race of the ball bearing **310** while the outer race does not rotate. The arms **312***a*, **312***b* are attached to the outer race so that they do not rotate with the friction stir weld backing **130**. The ball bearing **310** is positioned on the frictions stir weld backing **130** so that the arms **312***a*, **312***b* and the lower inert gas source **160** coupled thereto are positioned away from the lower surface **203** of the work piece **110**. The arms **312***a*, **312***b* can have extensions or fingers **314***a*, **314***b* that extend from the arms **312***a*, **312***b* towards the lower surface **203** of the work piece **110**. The fingers **314***a*, **314***b* can contact the lower surface **203** with minimal contact, such as the pointed nature of the fingers **314***a*, **314***b* shown in FIG. **3**. The minimal contact limits the friction between the fingers **314***a*, **314***b* and the lower surface **203** of the work piece **110** as the friction stir welding process traverses **206** along the work piece **110**. The friction that is generated can be sufficient enough to assist in preventing or minimizing the amount of rotation of the arms **312***a*, **312***b*, since the ball bearing **310** may not completely prevent the friction stir weld backing **130** from imparting rotation to the arms **312***a*, **312***b*. Additionally, the fingers **314***a*, **314***b* assist in maintaining a separation between the lower surface **203** and the arms **312***a*, **312***b*. This separation can be maintained to prevent the lower inert gas source **160** from inadvertently contacting the lower surface **203** as the friction stir welding process traverses **206** along the work piece **110**. Alternatively, different carriage designs can be used to couple the lower inert purge gas source **160** to the friction stir weld backing **130** so that the lower inert purge gas source **160** moves with the friction stir welding process as it traverses **206** along the work piece **110**.

(54) FIG. **4**A is a perspective view of an example friction stir welding system **400** and FIG. **4**B is a partial cut-away view of the example friction stir welding system **400**. The friction stir welding system **400** includes a friction stir weld head **120** and a friction stir weld backing **130** that compress

work pieces **110***a*, **110***b* along a pathway **119**. The friction stir weld head **120** and a friction stir weld backing **130** rotate and traverse **206** along the pathway to create a weld **118** that joins the two work pieces **110***a*, **110***b* together. The weld **118** is formed in an inert purge gas atmosphere that assists in minimizing or preventing the formation of oxides.

(55) An upper inert purge gas source **140** includes a housing **145** that assists in containing the inert purge gas **150** about the friction stir weld head **120** where the weld **118** is being formed. In the example of FIGS. **4**A-**4**B, the housing **145** is hemispherical with a large, open end positioned about the pathway **119** and above the upper surfaces **202***a*, **202***b* of the work pieces **110***a*, **110***b*. The smaller, closed end of the hemispherical housing, the end including an apex of the curve of the hemisphere, is positioned about the friction stir weld head **120**. The housing **145** is coupled to the friction stir weld head **120** so the inert gas atmosphere contained by the housing **145** moves with the friction stir weld head **120** during the friction stir welding process. In this example, the housing **145** is centered about the friction stir weld head **120** so that the inert gas atmosphere extends both ahead of and behind the friction stir weld head **120**. By extending behind the friction stir weld head **120**, the inert gas atmosphere contained by the housing **145** is maintained over the recently formed weld **118** for a period of time, such as while the weld **118** cools. Maintaining the inert purge gas atmosphere over the recently formed weld **118** can also assist in minimizing or preventing the formation of oxides, since oxides may be more readily formed in the recently formed weld **118** due to the residual heat in the work pieces **110***a*, **110***b* caused by the friction stir welding process. Allowing the recently formed weld **118** to cool, at least partially, in the inert purge gas atmosphere contained by the housing **145** can prevent the recently formed, oxide-susceptible weld **118** from being exposed to the normal environment. As the friction stir weld head **120** traverses **206** along the pathway **119**, the formed weld **118** will become exposed to the normal environment, but will have cooled to a sufficient level that the risk of oxide formation is significantly reduced from the risk of oxide formation without the inert gas environment.

(56) The housing **145** includes an inlet **410***a* through which the inert purge gas **150** can be dispensed into the housing **145**. The flow of inert purge gas **150** into the housing **145** can be substantially constant to maintain the inert purge gas atmosphere within the housing **145**. The inlet **410***a* can be coupled by tubing to the source of the inert purge gas **150** and the tubing can allow the housing **145** to move with the friction stir weld head **120** while maintaining the flow of inert purge gas **150** to the housing **145** during the friction stir welding process. Alternatively, the inert purge gas **150** can be introduced to the housing **145** prior to the friction stir welding process and the source of the inert purge gas **150** can then be disconnected from the housing **145**. In this manner, the inert purge gas atmosphere within the housing **145** is formed initially and additional inert purge gas **150** is not added during the friction stir welding process. This can allow the friction stir weld head **120** to move freely or easily since the housing **145** is not coupled the inert purge gas **150** source continuously during the friction stir welding process. In another alternative, the inlet **410***a* of the housing **145** can be connected to the inert purge gas source **150** during the friction stir welding process to replenish the inert purge gas atmosphere within the housing **145**.

(57) To assist with containing the inert purge gas **150** within the housing **145**, the housing **145** can include an interface that contacts the upper surfaces **202***a*, **202***b* of the work pieces **110***a*, **110***b* to assist in preventing the flow of inert purge gas **150** from the housing **145**. In an example, the interface can be a flexible skirt that extends from the perimeter of the large, open end of the housing **145** and contacts the upper surfaces **202***a*, **202***b*. The flexible nature of the skirt does not impede the traverse **206** of the friction stir weld head **120** and housing **145** and can reduce or minimize the flow of inert purge gas **150** from the housing **145**. In another example, the interface can be a brush strip or door sweep-like interface that includes densely packed fibers that extend from the perimeter of the large, open end of the housing **145** and contacts the upper surfaces **202***a*, **202***b*. The densely packed nature of the interface slows the escape of inert purge gas **150** from the housing **145**. In yet another example, a liquid interface can be used to prevent the flow of inert

purge gas **150** from the housing **145**. A liquid can be placed between the perimeter of the housing **145** and the upper surfaces **202***a*, **202***b*. The surface tension of the liquid can form a barrier between the perimeter of the housing **145** and the upper surfaces **202***a*, **202***b*, and can help contain the inert purge gas **150** within the housing **145**.

(58) In a further example, the interface can be made of a solid material that is less hard than the work pieces **110***a*, **110***b* and has a low coefficient of friction. The low coefficient of friction will allow the interface to move easily across the upper surfaces **202***a*, **202***b* without impeding the traverse **206** of the friction stir weld head **120** and the housing **145**. The lower hardness of the interface prevents it from scratching or damaging the upper surfaces **202***a*, **202***b*, allowing the interface to directly contact the upper surfaces **202***a*, **202***b* and minimize the separation between the interface and upper surfaces **202***a*, **202***b*. By minimizing the separation, the area through which inert purge gas **150** can flow from the housing **145** is reduced. While any design of the interface may allow some portion of the inert purge gas **150** to escape the housing **145**, the constant flow of inert purge gas **150** into the housing **145** prevents the normal atmosphere from entering the housing **145**, which maintains the inert purge gas atmosphere about the friction stir welding process.

(59) The housing **145** can be coupled directly to the friction stir weld head **120** or can be attached to a carriage that moves with the friction stir weld head **120**, such as a carriage coupled to the friction stir weld head **120** or a separate carriage that moves with the friction stir weld head **120**. In an embodiment in which the housing **145** is coupled directly to the friction stir weld head **120** or to a carriage of the friction stir weld head **120**, the housing **145** can be coupled in a manner so that the housing **145** does not rotate with the friction stir weld head **120**. For example, the closed end housing can be coupled to a portion of the friction stir weld head **120** that does not rotate. In another example, the housing **145** can be coupled to a carriage of the frictions stir weld head **120**, such as a ball bearing similar to **310** of FIG. **3**. The friction stir weld head **120** can include a ball bearing, with the inner race of the bearing rotating with the friction stir weld head **120**, while the outer race is coupled to the housing **145** and does not substantially rotate with the friction stir weld head **120**.

(60) A housing **165**, similar to the housing **145**, can be positioned beneath the work pieces **110***a*, **110***b* to form an inert purge gas atmosphere about the lower surfaces **203***a*, **203***b* during the friction stir welding process. Like the housing **145**, the housing **165** can be hemispherical and can contain an inert purge gas atmosphere about the friction stir weld backing **130**. The housing **165** can similarly include an interface between the housing **145** and the lower surfaces **203***a*, **203***b*. Additionally, the housing **165** can include an inlet **410***b* to flow inert purge gas **150** into the housing **165**. The housing **165** can move with the friction stir weld backing **130** during the friction stir welding process and can be coupled directly to the friction stir weld backing **130** or a carriage, similar to the housing **145**.

(61) In the example of FIGS. **4**A-**4**B, the housings **145**, **165** are hemispherical in shape and centered about the friction stir weld head **120** and friction stir weld backing **130**, respectively. In alternative embodiments, the housing **145**, **165** can be similarly shaped, i.e. both can have the same shape, but have a shape other than a hemispherical shape. For example, the housings **145**, **165** can be semi-oviod, rectangle, square or other shaped. The shape can be selected based on a variety of factors, such as the size of the weld **118** being formed, the duration which the weld **118** should remain in the inert gas atmosphere after being formed, the shape or profile of the work pieces **110***a*, **110***b*, other considerations, or combinations thereof. In an embodiment, the housings **145**, **165** can be semi-rigid or flexible to allow them to conform to the profile of the work pieces **110***a*, **110***b*, such as to allow the friction stir welding process to occur on work pieces having a curved profile. In another embodiment, the housings **145**, **165** can be rigid. Additionally, the housings **145** and **165** can be dissimilar, such as each having a different shape, size or other characteristics.

(62) FIG. **5** is an example friction stir welding method **500**. The method **500** includes forming an inert purge gas atmosphere in which a friction stir welding process will occur. By performing the

friction stir welding process in the inert purge gas atmosphere, the formation of oxides due to the friction stir welding process will be reduced or minimized. The formation of oxides in the work pieces joined by the friction stir welding process can weaken the weld and cause it to mechanically fail.
(63) At **502**, an inert purge gas atmosphere is formed about an upper portion of a pathway. The pathway is the abutment of the two work pieces, along which the friction stir welding process will occur to join the two work pieces. The inert purge gas atmosphere is formed about the pathway at the upper surfaces of the two work pieces, which are the surfaces that are contacted by an upper shoulder of a friction stir weld head.
(64) At **504**, an inert purge gas atmosphere is, optionally, formed about a lower portion of the pathway. The inert purge gas atmosphere is formed about the pathway at the lower surfaces of the two work pieces, which are the surfaces that are contacted by a friction stir weld backing. The combination of **502** and **504** forms an inert purge gas atmosphere about the upper and lower portions of the pathway, which assists in reducing the formation of oxides caused by the friction stir welding process.
(65) At **506**, a friction stir welding process is performed within the inert purge gas atmosphere(s). As previously discussed, performing the friction stir welding process in such a manner assists in minimizing or preventing oxide formation as a result of the friction stir welding process. By reducing the formation of oxides, higher quality welds can be formed by the friction stir welding process. Additionally, the tolerances of the friction stir welding process, such as the rotational and traverse speeds of the friction stir weld head, are expanded while a similar quality of weld is maintained. This can allow the friction stir welding process to be sped up while maintaining a good quality of weld formation. Further, the inert purge gas atmosphere can allow some materials to be friction stir welded, such as high lithium content Al—Li alloys. High lithium content Al—Li alloys can be prone to oxide formation, making them unsuitable to be joined using friction stir welding. However, the inert purge gas atmosphere assists in minimizing or preventing oxide formation, which allows such high lithium content Al—Li alloys to be joined using friction stir welding.
(66) The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the disclosure. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the systems and methods described herein. The foregoing descriptions of specific embodiments or examples are presented by way of examples for purposes of illustration and description. They are not intended to be exhaustive of or to limit this disclosure to the precise forms described. Many modifications and variations are possible in view of the above teachings. The embodiments or examples are shown and described in order to best explain the principles of this disclosure and practical applications, to thereby enable others skilled in the art to best utilize this disclosure and various embodiments or examples with various modifications as are suited to the particular use contemplated. It is intended that the scope of this disclosure be defined by the following claims and their equivalents.

Claims

1. A friction stir welding system, comprising: a friction stir weld head on a first side of multiple work pieces to join the multiple work pieces along a weld pathway; a friction stir weld backing spaced apart from the friction stir weld head by at least a portion of the multiple work pieces, the friction stir weld backing on a second side of the multiple work pieces opposite the first side of the multiple work pieces; an inert purge gas source to output inert purge gas on the first side of the multiple work pieces; a housing configured to at least partially contain the inert purge gas about the weld pathway on the first side of the multiple work pieces, wherein the inert purge gas source is located so as to output the inert purge gas between the housing and the first side of the multiple work pieces; a second inert purge gas source to output the inert purge gas on the second side of the

multiple work pieces; and a second housing configured to at least partially contain the inert purge gas about the weld pathway on the second side of the multiple work pieces, wherein the second inert purge gas source is located so as to output the inert purge gas between the second housing and the second side of the multiple work pieces.

2. The friction stir welding system of claim 1, wherein the inert purge gas source comprises an inlet on the housing and facing the first side of the multiple work pieces.

3. The friction stir welding system of claim 1, wherein the housing is coupled directly to the friction stir weld head via ball bearings.

4. The friction stir welding system of claim 1, wherein the housing is configured to move with the friction stir weld head during a friction stir welding process.

5. The friction stir welding system of claim 1, wherein the housing extends both ahead and behind the friction stir weld head so as to be configured to maintain, for a period of time, an inert purge gas atmosphere over a recently formed weld produced by the friction stir weld head.

6. The friction stir welding system of claim 1, wherein the inert purge gas source comprises tubing to allow the housing to move with the friction stir weld head while maintaining flow of the inert purge gas to the housing during a friction stir welding process.

7. The friction stir welding system of claim 1, further comprising a skirt that is flexible so as to be configured to conform to curved profiles of the multiple work pieces, wherein the flexible skirt is attached to the housing and extends from a perimeter of the housing, and the flexible skirt is configured to contact the first side of the multiple work pieces.

8. The friction stir welding system of claim 1, wherein the housing comprises a brush that extends from a perimeter of the housing and is configured to contact the first side of the multiple work pieces.

9. The friction stir welding system of claim 1, wherein the housing comprises a liquid interface that extends from a perimeter of the housing and is configured to contact the first side of the multiple work pieces.

10. The friction stir welding system of claim 1, wherein the housing includes an interface that extends from a perimeter of the housing and is configured to contact the first side of the multiple work pieces, wherein the interface comprises a solid material that is less hard than the multiple work pieces.

11. The friction stir welding system of claim 1, wherein the housing is semi-rigid or flexible so as to be configured to conform to profiles of the multiple work pieces.

12. The friction stir welding system of claim 11, wherein the profiles of the multiple work pieces are curved.

13. A friction stir welding system, comprising: a friction stir weld head configured to be on a top side of multiple work pieces to join the multiple work pieces along a weld pathway; an inert purge gas source configured to output inert purge gas on the top side of the multiple work pieces; a housing configured to at least partially contain the inert purge gas about the weld pathway on the top side of the multiple work pieces, wherein the inert purge gas source is located so as to output the inert purge gas between the housing and the top side of the multiple work pieces, and wherein the housing is flexible so as to be configured to conform to curved profiles of the multiple work pieces; a second inert purge gas source configured to output the inert purge gas on a bottom side of the multiple work pieces; and a second housing configured to at least partially contain the inert purge gas about the weld pathway on the bottom side of the multiple work pieces, wherein the second inert purge gas source is located so as to output the inert purge gas between the second housing and the bottom side of the multiple work pieces.

14. The friction stir welding system of claim 13, wherein the housing is coupled directly to the friction stir weld head via ball bearings.

15. The friction stir welding system of claim 13, wherein the housing is configured to move with the friction stir weld head during a friction stir welding process.

16. The friction stir welding system of claim 13, further comprising a friction stir weld backing spaced apart from the friction stir weld head by at least a portion of the multiple work pieces, the friction stir weld backing on a bottom side of the multiple work pieces opposite the top side of the multiple work pieces.
17. The friction stir welding system of claim 13, wherein the housing extends both ahead and behind the friction stir weld head so as to be configured to maintain, for a period of time, an inert purge gas atmosphere over a recently formed weld produced by the friction stir weld head.
18. A friction stir welding system, comprising: a friction stir weld head on a top side of multiple work pieces to join the multiple work pieces along a weld pathway; a friction stir weld backing spaced apart from the friction stir weld head by at least a portion of the multiple work pieces, the friction stir weld backing on a bottom side of the multiple work pieces opposite the top side of the multiple work pieces; an inert purge gas source to output inert purge gas on the top side of the multiple work pieces; a housing configured to at least partially contain the inert purge gas about the weld pathway on the top side of the multiple work pieces, wherein the inert purge gas source is located so as to output the inert purge gas between the housing and the top side of the multiple work pieces; a second inert purge gas source to output the inert purge gas i) on the bottom side of the multiple work pieces and ii) on sides of the friction stir weld backing; and a second housing configured to at least partially contain the inert purge gas about the weld pathway on the bottom side of the multiple work pieces, wherein the second inert purge gas source is located so as to output the inert purge gas between the second housing and the bottom side of the multiple work pieces.